

PAPER198: FUBii Taxonomy Part 1 — Compact Object and Stellar Physics Buoyancy Forces

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$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \text{Bigl}(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2}) \cdot \text{Bigr}, \quad [SSq] = 0.57$$

<!-- ? = 5.0e-4 day⁻¹, [SSq] = 0.57, B_i = 6.1e-1 -->

Abstract

The UQFF buoyancy force *FUBii* is applied across all major astrophysical phenomena by embedding each system's characteristic energy or length scale into the universal Frel/ELEP scaling framework. This paper catalogs 18 unique FUBii variants relating to compact objects and stellar physics: MHD dynamo buoyancy, terminal velocity, Press-Schechter, Hawking radiation, quasi-normal mode ringdown, Blandford-Znajek jet power, Arnett supernova, entanglement, jet velocity, planet migration, AGN feedback, angular momentum accretion, J-type shock, Sedov-Taylor blast wave, GRB afterglow, SIDM core formation, ionization fronts, superfluid glitch, and virial theorem.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. General FUBii Framework

$$F_{UBii,X} = \pm F_{rel} \times (F_X / E_{LEP}) \times Q_{wave} \times [\text{decay/oscillation factor}]$$

where:

$F_{rel} \sim 4.3 \times 10^{33} \text{ N}$ (relativistic UQFF force scale)

E_{LEP} = energy of characteristic LEP scale (J)

Q_{wave} = quantum wave amplitude (J/m³), std $\sim 6.33 \times 10^4$

F_X = system-specific physical expression

2. Compact Object Variants

2.1 MHD Dynamo Buoyancy

$$F_{UBii,MHD} = F_{rel} \times (E_M/t_{eddy}/E_{LEP}) \times Q_{wave} \times (3/2)(Re_m^{1/2} - 1)$$

E_M = magnetic energy density

$t_{\text{eddy}} = l/v \sim \text{Myr}$ (eddy turnover time)
 $Re_m \sim 10^{10}$ (magnetic Reynolds number)
Source: BB_C_Equations item 748

2.2 Terminal Velocity Buoyancy

$F_{\text{UBii,termv}} = F_{\text{rel}} \times (v(2GM(1-G)/r_{\text{launch}}) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (t \cdot L/c)$
 $G = L/L_{\text{Edd}} \sim 1$ (Eddington ratio for wind-driven systems)
 $r_{\text{launch}} \sim 100 R_s$ (wind launch radius)
 t = optical depth
Source: BB_C_Equations item 853, 1823

2.3 Hawking Radiation Buoyancy

$F_{\text{UBii,haw}} = F_{\text{rel}} \times (hc^3/(8pGMk_B) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (?(2p))$
 $T_H = hc^3/(8pGMk_B)$ (Hawking temperature)
 $? = c^4/(4GM)$ (surface gravity)
Source: BB_C_Equations item 901, 1246

2.4 Quasi-Normal Mode Ringdown Buoyancy

$F_{\text{UBii,qnm}} = -F_{\text{rel}} \times (c^3/(2pGM) \cdot (0.3737 + 0.088 \cdot a_f) / E_{\text{LEP}})$
 $\times Q_{\text{wave}} \times e^{-t/t} \times [t ? M]$
 $a_f \sim 0.69$ (dimensionless BH spin post-merger)
 t = ringdown decay time
 $f_{\text{QNM}} = c^3/(2pGM_f) \cdot f(0.3737 + 0.088 \cdot a_f)$ (Berti fits $l=2, m=2$ mode)
Source: BB_C_Equations item 945, 1293

2.5 Blandford-Znajek Jet Power Buoyancy

$F_{\text{UBii,bz}} = F_{\text{rel}} \times (1/32 \cdot B^2 R_{\text{H40}} H^2/c / E_{\text{LEP}})$
 $\times Q_{\text{wave}} \times (ac/(2R_H)) \times (1+t_{\text{var}})$
 $R_H = GM/c^2$ (horizon radius)
 $\Omega_H = ac/(2R_H)$ (BH angular velocity)
Source: BB_C_Equations item 1147
 $F_{\text{UBii,bz2}} = F_{\text{rel}} \times (?(16p) \cdot F_{\text{BH}}^2 \cdot \Omega_{\text{BH}}^2/c / E_{\text{LEP}})$
 $\times Q_{\text{wave}} \times (ac^3/(2GM)) \times 0.05^{-1}$
 $F_{\text{BH}} = B \cdot p \cdot r_{\text{H}}^2$ (BH magnetic flux thread; EHT-calibrated)
Source: BB_C_Equations item 967, 1316

2.6 Arnett Supernova Buoyancy

$F_{\text{UBii,arnett}} = F_{\text{rel}} \times (M_{\text{Ni}} \cdot e_{\text{Ni}}/t_d / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-t/t}$
 M_{Ni} = nickel mass ($\sim 0.5 M_{\text{?}}$)
 e_{Ni} = nickel decay energy
 $t_d^2 = 3?M/(4pcv^2)$ (diffusion time)
Source: BB_C_Equations item 1035

2.7 TOV Hydrostatic Equilibrium Buoyancy

$F_{\text{UBii,tov}} = -F_{\text{rel}} \times (dP/dr = -Gm(r)?(r)/r^2 \cdot (1 + P(r)/(?(r)c^2))$
 $\times (1 + 4pr^3P(r)/(m(r)c^2)) / (1 - 2Gm(r)/(rc^2)) / E_{\text{LEP}}) \times Q_{\text{wave}}$
Includes all GR corrections (Schwarzschild metric, pressure-energy coupling)
Source: BB_C_Equations item 1300

2.8 Pulsar Spin-Down Buoyancy

$F_{\text{UBii,puls}} = -F_{\text{rel}} \times (t = P/(2?) / E_{\text{LEP}}) \times Q_{\text{wave}}$

P = period, $\dot{P} \sim 10^{15}$ s/s (period derivative, Vela)
 $I \cdot d\Omega/dt = -L/\Omega$ (torque equation)
Source: BB_C_Equations item 1302

3. Stellar/Accretion Variants

3.1 Jet Velocity Buoyancy

$F_{UBii,jetvel} = F_{rel} \times (v_j \sim v_K \cdot (r_A/r_0)^{1/2} / E_{LEP})$
 $\times Q_{wave} \times (B/v(4\pi)) \times (1+t/t_A)$
 $v_K = v(GM/r_0)$ (Keplerian velocity at footpoint, $r_0 = 1-10$ AU)
 r_A = Alfvén radius (10-50 AU, from POETS protostellar data)
Source: BB_C_Equations item 1096, 1272

3.2 Type-I Planet Migration Buoyancy

$F_{UBii,migr} = -F_{rel} \times (-f \cdot 2(GM_p)^2 \cdot S/(M_* \cdot 0.05 \cdot (H/r)^3) / E_{LEP})$
 $\times Q_{wave} \times [t = M_p/\dot{M}_{acc} \sim 10^6 \text{ yr}]$
 $f \sim -1.36$ (Lindblad/corotation factor, inward migration)
 S = disk surface density
Source: BB_C_Equations item 1121, 1322

3.3 Superfluid Vortex Glitch Buoyancy

$F_{UBii,glitch} = F_{rel} \times (I_s/I \cdot \dot{\Omega} \cdot (1-e^{-t/t_q}) / E_{LEP})$
 $\times Q_{wave} \times \dot{\Omega} \times e^{-t/t_q}$
 I_s = superfluid moment of inertia ($\sim 10^{38}$ kg·m²)
 t_q = quench timescale
 $\dot{\Omega}$ = angular velocity jump
Source: BB_C_Equations item 1753, 1304

4. Shock and Blast Wave Variants

4.1 J-Type Shock Buoyancy (Rankine-Hugoniot)

$F_{UBii,jshock} = F_{rel} \times ((\gamma v_1^2 + P_1) / E_{LEP})$
 $\times Q_{wave} \times (v_2/v_1) \times (\gamma+1)/(\gamma-1+2/M^2)$
 $\gamma = 5/3$ (adiabatic index)
 $M = v_1/c_s$ (Mach number, from Chandra X-rays in HH 154)
Source: BB_C_Equations item 1193, 1274

4.2 Sedov-Taylor Blast Wave Buoyancy

$F_{UBii,sedov} = F_{rel} \times ((E \cdot t^2/\gamma)^{1/5} / E_{LEP}) \times Q_{wave} \times [d/dt(Mv)=0] \times t^{2/5}$
 $E \sim 10^{51}$ erg (explosion energy)
 γ = ambient density
 $R(t) = (Et^2/\gamma)^{1/5}$ (self-similar blast radius)
Source: BB_C_Equations item 1207, 1288

4.3 C-Type Shock Buoyancy (Magnetized)

$F_{UBii,cshock} = -F_{rel} \times ((\gamma \times B) \times B/(4\pi) + \gamma_i \cdot \nu_{in} \cdot (v_n - v_i) / E_{LEP}) \times Q_{wave}$
C-shocks: continuous, multi-fluid MHD (ions+neutrals+magnetic field coupled)
 ν_{in} = ion-neutral collision frequency
Source: BB_C_Equations item 1276

5. Feedback and Outflow Variants

5.1 AGN Feedback Buoyancy

$$F_{UBii,agn} = F_{rel} \times (f(v_{out}) \cdot L_{AGN}/c / E_{LEP}) \times Q_{wave} \times (?_{out} \cdot v_{out}) \times v_{out}^{?1}$$

$$\text{Momentum-driven: } p_{term} = ?_{out} \cdot v_{out} = f(v_{out}) \cdot L_{AGN}/c$$

$$?_{out} \sim 10\text{--}100 \text{ } M_{\odot}/\text{yr}$$

Source: BB_C_Equations item 1165, 1314

5.2 Angular Momentum Accretion Buoyancy

$$F_{UBii,ang} = -F_{rel} \times (? \cdot r^2 \cdot 0 / E_{LEP}) \times Q_{wave} \times T_B \times e^{-t/t_{disk}}$$

$$T_B = B^2 r^3 / (4p) \text{ (magnetic braking torque)}$$

$$t_{disk} = \text{disk decay timescale}$$

Source: BB_C_Equations item 1189, 1270

5.3 Feedback Energy Coupling Buoyancy

$$F_{UBii,coup} = F_{rel} \times ((1/2) \cdot ?_w \cdot v_w^2 / (?_{acc} \cdot c^2 \cdot 10) / E_{LEP}) \times Q_{wave} \times 0.05\text{--}0.1$$

$$e_f = E_{kin} / (?_{acc} \cdot c^2) \sim 0.05\text{--}0.1 \text{ (coupling fraction)}$$

Source: BB_C_Equations item 1331, 1554

6. Structure Formation Variants

6.1 Press-Schechter Halo Mass Function Buoyancy

$$F_{UBii,ps} = F_{rel} \times (dn/dM = v(2/p) \cdot (?0/M) \cdot (d_c/s(M)) \cdot |d \ln s / d \ln M| \cdot \exp(-d^2_c/(2s^2))) / E_{LEP}$$

$$\times Q_{wave} \times ?0 \text{ (Gaussian part from } s \sim d_c)$$

$$d_c \sim 1.69 \text{ (critical collapse overdensity)}$$

Source: BB_C_Equations item 877, 1574

6.2 Structure Growth Rate Buoyancy

$$F_{UBii,grow} = -F_{rel} \times (d'' + 2H \cdot d' = (3/2) \cdot 0_m \cdot H^2 \cdot d/a^3 / E_{LEP}) \times Q_{wave}$$

$$D(a) = 50_m/2 \cdot ?0^a da' / (a'H(a')/H0)^3 \text{ (growth factor)}$$

$$\text{Growing mode: } D \propto a \text{ in matter era, suppressed by dark energy}$$

Source: BB_C_Equations item 1371, 1244

7. GRB Variants

7.1 GRB Fireball Expansion Buoyancy

$$F_{UBii,fire} = F_{rel} \times (G(r) = r/R0 \text{ } (r < R_s), G = ? \text{ } (r > R_s) / E_{LEP}) \times Q_{wave}$$

$$R0 \sim 107 \text{ cm (initial fireball radius)}$$

$$R_s = ?^2 R0 \text{ (saturation radius)}$$

Source: BB_C_Equations item 1482, 1306

7.2 GRB Afterglow Synchrotron Buoyancy

$$F_{UBii,after} = -F_{rel} \times (F_{\gamma} \propto t^{-p} \cdot t^{-3(p-1)/4} (?_m < ?_c) / E_{LEP}) \times Q_{wave}$$

$$p \sim 2.2\text{--}2.5 \text{ (electron power-law index)}$$

$$\text{Electrons accelerated by DSA, spectrum } N(E) \propto E^{-p}$$

Source: BB_C_Equations item 1227, 1308

8. References

grok_share_7514fe.txt lines 2443–2680 (BBCEquations_04Sept2025.pdf Part 1 catalogue)

PAPER_196: Triadic Master Equation System

PAPER197: *FUBii* Extended Integral